

ORIGINAL RESEARCH

Combined LDL-C, Lp(a), and hsCRP Assessment Predicts ASCVD in the Multiethnic HELIUS Cohort

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ABSTRACT

BACKGROUND Low-density lipoprotein cholesterol (LDL-C), lipoprotein(a) (Lp[a]), and high-sensitivity C-reactive protein (hsCRP) capture different pathways driving atherosclerotic cardiovascular disease (ASCVD). However, it is unknown whether their combined assessment provides reliable risk stratification in different ethnic populations with markedly different distributions of these biomarkers as well as background risk of ASCVD.

OBJECTIVES The goal was to evaluate independent and joint associations of LDL-C, Lp(a), and hsCRP with incident ASCVD and to formally assess additive and multiplicative interaction of these biomarkers in a multiethnic cohort, and to determine whether these associations are consistent across participants of African, European, and South Asian Surinamese descent.

METHODS Among 15,676 HELIUS (HEalthy Life In an Urban Setting) participants without prior myocardial infarction or ischemic stroke (mean age 44.4 ± 13.2 years; 56.0% women), incident ASCVD was identified through linkage to nationwide registries. Fine-Gray competing-risk models were used to assess associations across increasing biomarker quintiles and joint elevations. Additive interaction was assessed using the relative excess risk due to interaction, attributable proportion, and synergy index. Incremental predictive value of Lp(a) and hsCRP beyond a PREVENT-ASCVD (Predicting Risk of Cardiovascular Disease EVENTS-Atherosclerotic Cardiovascular Disease)-based clinical model was evaluated using time-dependent area under the curve and continuous net reclassification improvement.

RESULTS Over median 8.9-year follow-up (138,375 person-years), 378 ASCVD events occurred. Top vs bottom quintile adjusted subdistribution HRs were 1.39 (95% CI: 1.09-1.79) for LDL-C, 1.86 (95% CI: 1.54-2.25) for Lp(a), and 1.51 (95% CI: 1.34-1.71) for hsCRP. Formal interaction assessment showed no interaction on either the multiplicative or additive scale. Risk increased with the number of elevated biomarkers, reaching a 2.44-fold increase (95% CI: 1.46-4.09) among those with all 3 elevated. Consistent dose-response patterns were observed across ethnic groups ($P_{\text{interaction}} = 0.75$), although absolute ASCVD risk was approximately 3-fold higher in South Asian Surinamese participants. Addition of Lp(a) and hsCRP to a clinical model modestly improved discrimination at 10 years (Δ area under the curve: 0.006; $P = 0.030$) and reclassification (continuous net reclassification improvement: 0.165; 95% CI: 0.050-0.289).

CONCLUSIONS In a multiethnic primary prevention cohort, combined assessment of LDL-C, Lp(a), and hsCRP identified individuals at elevated long-term risk through independent pathways, with consistent relative risk gradients but divergent absolute risks across ethnicities. (JACC. 2026;■:■-■) © 2026 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

ABBREVIATIONS
AND ACRONYMS

apoB = apolipoprotein B

ASCVD = atherosclerotic
cardiovascular diseaseaSHR = adjusted
subdistribution hazard ratios

AUC = area under the curve

HDL-C = high-density
lipoprotein cholesterolhsCRP = high-sensitivity C-
reactive proteinLDL-C = low-density
lipoprotein cholesterol

Lp(a) = lipoprotein(a)

NRI = net reclassification
improvementSHR = subdistribution hazard
ratio

Atherosclerotic cardiovascular disease (ASCVD) is driven by lipid accumulation and arterial inflammation. Apolipoprotein B (apoB)-containing lipoproteins, such as low-density lipoprotein cholesterol (LDL-C) and lipoprotein(a) (Lp[a]), are causal for atherogenesis.¹⁻³ These lipoproteins promote plaque formation and trigger inflammatory signaling that contributes to plaque progression and rupture. A major inflammatory pathway involves interleukin-1 signaling, with high-sensitivity C-reactive protein (hsCRP) as a downstream marker.⁴ Although hsCRP is not itself causal,⁵ it is consistently associated with ASCVD events.⁶⁻⁸

Recent primary prevention studies have demonstrated that LDL-C, Lp(a), and hsCRP are each independently associated with ASCVD and that joint elevations identify individuals at particularly high long-term risk.⁹⁻¹¹ However, these studies (WHS [Women's Health Study], EPIC-Norfolk [European Prospective Investigation into Cancer and Nutrition in Norfolk], and UK Biobank) were carried out in populations of almost exclusively European ancestry. This leaves a critical gap: neither lipid-related nor inflammatory risk is uniformly distributed across different ethnic populations. Individuals of African and South Asian descent have markedly higher circulating concentrations of Lp(a) and hsCRP,¹²⁻¹⁶ while also experiencing a higher burden of ASCVD than individuals of European descent.^{17,18} It is unknown whether the conclusions from previous investigations into this biomarker triad are preserved across diverse populations in which baseline biomarkers and ASCVD incidence are fundamentally different.

A second unresolved question concerns the nature of the interaction among LDL-C, Lp(a), and hsCRP. Prior studies have reported mixed findings regarding multiplicative interaction between these biomarkers,^{11,19-24} but none have formally assessed additive interaction, which carries distinct implications for joint risk assessment. If ASCVD risk mediated by

these biomarkers accumulates additively, patients with multiple elevations may benefit from concurrent interventions targeting these pathways, such as lipid-lowering therapy, emerging Lp(a)-lowering agents, and anti-inflammatory drugs such as colchicine.²⁵⁻²⁷

Therefore, we examined the independent and joint associations of LDL-C, Lp(a), and hsCRP with incident ASCVD in a large, multiethnic population-based cohort, among individuals without prior ASCVD. In addition, we evaluated whether ASCVD risk associated with cumulative biomarker burden is consistent across participants of African, European, and South Asian Surinamese descent in the cohort. Finally, we formally tested for additive and multiplicative interaction, and determined the incremental predictive value of Lp(a) and hsCRP beyond a clinical risk model.

METHODS

STUDY POPULATION AND DESIGN. The HELIUS (HEalthy Life In an Urban Setting) study is an ongoing, multiethnic, population-based cohort conducted in Amsterdam, the Netherlands, details of which have been published previously.^{28,29} A total of 24,780 individuals of Dutch, South Asian Surinamese, African Surinamese, Ghanaian, Turkish, and Moroccan descent, aged 18 to 70 years, were enrolled between 2011 and 2015. Ethnicity was classified based on the country of birth: participants born outside the Netherlands with at least 1 parent born in the same country and those born in the Netherlands with both parents born abroad were considered non-Dutch. The distinction between African Surinamese and South Asian Surinamese was determined by self-report. The African Surinamese population is largely descended from West Africans who were brought to Suriname as part of the transatlantic slave trade between the 17th and 19th centuries. The South Asian Surinamese population is descended from labor migrants who migrated from British-ruled India to Suriname at the end of the 19th and the beginning of the 20th century. Following the independence of

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

Suriname from the Netherlands in 1975, a substantial number of African Surinamese and South Asian Surinamese individuals migrated to the Netherlands.

Data collection involved a physical examination, completion of a questionnaire, and the collection of biological samples. The HELIUS study was approved by the Ethical Review Board of Amsterdam UMC (MREC 10/100# 17.10.1729) and is conducted in accordance with the principles of the Declaration of Helsinki. All participants provided written informed consent. Participants were included in the present study if baseline measurements for Lp(a), hsCRP, and LDL-C were available and if written consent was given for linkage of their study data to health registries. Participants who reported myocardial infarction or ischemic stroke at baseline were excluded from the analyses.

LABORATORY MEASUREMENTS. Fasting plasma samples were collected at baseline and stored at -80°C . Total cholesterol, high-density lipoprotein cholesterol (HDL-C), and triglycerides were measured by colorimetric spectrophotometry (Roche Diagnostics). LDL-C was calculated using the Friedewald formula. Lp(a) concentrations were measured using the Roche Tina-Quant Lipoprotein(a) immunoturbidimetric molar assay (Roche Diagnostics), which is standardized to be largely independent of apolipoprotein(a) isoform size. hsCRP concentrations were measured turbidimetrically with a Cobas 702c analyzer (Roche Diagnostics). If the measured hsCRP concentration was below the lower limit of detection (<0.3 mg/L), the value was replaced by 0.15 mg/L ($n = 1,377$). Missingness of biomarkers at baseline was low overall ($<1\%$ for each) (Supplemental Figure 1).

OUTCOME DEFINITIONS AND LINKAGE. Study data were linked to mortality data and hospital admission records from Statistics Netherlands.^{30,31} Mortality data included all registered deaths and causes of death from 2011 to 2023 registered in the Cause of Death Registry. Hospital admissions records from Dutch Hospital Data included all admissions (≥ 1 day) to general and academic hospitals in the Netherlands from 2011 to 2022. After linkage, the records were pseudonymized by Statistics Netherlands and made available for analysis. Events were selected according to the International Classification of Diseases, 9th Revision, for events occurring in 2011 and 2012 and 10th Revision for events occurring from 2013 to 2023. The primary outcome was a composite of nonfatal myocardial infarction, nonfatal ischemic stroke, coronary revascularization, and cardiovascular mortality (Supplemental Table 1). Incidence of coronary

revascularization, defined as percutaneous coronary intervention or coronary artery bypass grafting, was assessed using Dutch national procedure codes.

STATISTICAL ANALYSES. Continuous variables are presented as mean $\pm$ SD or median (IQR), as appropriate; categorical variables are reported as counts and percentages. Spearman correlation coefficients were calculated between biomarker pairs. To enhance statistical power, the 6 ethnicities were combined into 3 groups based on overlapping genetic composition and geographical origins, as described previously.³² Dutch and Turkish participants were classified as European; African Surinamese, Ghanaian, and Moroccan participants were classified as African; and South Asian Surinamese participants were classified as a distinct group.

Associations between biomarker quintiles and incident ASCVD were assessed in a time-to-first-event analysis using Fine-Gray subdistribution hazard models, with noncardiovascular death as a competing event. Follow-up was calculated from the baseline visit to the first ASCVD event, death, or the last available follow-up date. The first event was used for the analysis in cases in which multiple events occurred in a single participant. Subdistribution hazard ratios (SHRs) were estimated for each quintile relative to the lowest. Minimally adjusted models included age, sex, ethnicity, and the 2 other biomarkers as continuous covariates. Fully adjusted models additionally included current smoking, lipid-lowering therapy, systolic blood pressure, diabetes mellitus, estimated glomerular filtration rate, and microalbuminuria. Linear trends were tested by modeling biomarker quintiles as ordinal variables. As sensitivity analyses, biomarkers also were modeled continuously (LDL-C untransformed; Lp[a] and hsCRP natural log-transformed) and using established clinical cutoffs: hsCRP (<1 , $1-3$, ≥ 3 mg/L), Lp(a) (<75 , $75-125$, ≥ 125 nmol/L), and LDL-C (<1.8 , $1.8-3.5$, ≥ 3.5 mmol/L).

To evaluate joint associations, participants were cross-classified as high (top quintile) vs low (quintiles 1-4) status for each biomarker pair and for all 3 biomarkers simultaneously, with both (or all 3) biomarkers low as the reference. Multiplicative interaction was tested by comparing fully adjusted Fine-Gray models containing biomarkers transformed as restricted cubic splines against models additionally including spline interaction terms, using the likelihood ratio test. For the 3-way analysis, a model with all biomarker spline terms and pairwise product terms was compared against a model additionally including the 3-way product term.

The observed-to-expected joint SHR ratio was computed, with the expected joint SHR defined as the product of individual SHRs under multiplicative independence. Additive interaction was assessed using the relative excess risk due to interaction, the attributable proportion due to interaction, and the synergy index, with null values of 0, 0, and 1, respectively. CIs were derived using the delta method from the variance-covariance matrix of the Fine-Gray model coefficients. These interaction analyses were repeated using clinical cutoffs to define high biomarker status.

The cumulative burden of elevated biomarkers was categorized as 0, 1, or ≥ 2 biomarkers in the top quintile. Cumulative incidence of ASCVD was estimated at 5 and 10 years using the Aalen-Johansen estimator, stratified by biomarker burden category and ethnic group, with 95% CIs derived from the variance of the cumulative incidence function. Absolute risk differences at 10 years were calculated between each biomarker burden category and the reference (0 elevated biomarkers). Effect modification by ethnicity was tested using a likelihood ratio test comparing models with and without an interaction term between biomarker burden and ethnicity.

To evaluate whether Lp(a) and hsCRP improve risk prediction beyond established clinical models, we compared a clinical model with an enhanced model. The clinical model included the PREVENT-ASCVD (Predicting Risk of Cardiovascular Disease EVENTS-Atherosclerotic Cardiovascular Disease)-derived 10-year ASCVD risk score, which incorporates non-HDL-C (thereby capturing LDL-C along with other apoB-containing lipoproteins),^{33,34} together with the covariates specified previously. The enhanced model additionally included log-transformed Lp(a) and log-transformed hsCRP. Time-dependent receiver operating characteristic curves were constructed at 5 and 10 years using inverse probability of censoring weighting, accounting for competing risks, and the area under the receiver operating characteristic curve (AUC) was compared between models. To quantify overfitting, we additionally applied Harrell's bootstrap optimism correction (1,000 resamples) to the AUCs of both models; we report apparent and optimism-corrected estimates. Continuous net reclassification improvement (NRI) was estimated at 5 and 10 years using Kaplan-Meier methods with 1,000 bootstrap iterations for 95th percentile-based CIs, decomposed into NRI+ (events) and NRI- (non-events). Category-based reclassification was assessed at 10 years using American College of Cardiology/American Heart Association guideline-concordant risk thresholds (<3%, 3%-5%, 5%-10%, $\geq 10\%$).³⁴

Both models were fitted and evaluated on the same cohort; bootstrap optimism correction provided internal validation, but external validation was not performed. Exploratory ethnicity-stratified analyses assessed consistency of incremental predictive value across groups.

In all models that included ethnicity as a covariate, the 3-level aggregated ethnicity variable was included as a covariate, and the original 6-ethnicity variable was used as the clustering variable in these models. Cluster-robust standard errors were estimated using the original 6 ethnicity groupings as the clustering variable to account for potential within-group correlations. To assess potential heterogeneity within the aggregated ethnicities, likelihood ratio tests were performed within the African and European subgroups, with an interaction term between the independent variable of interest and the original 6-ethnicity variable. No statistically significant heterogeneity was found in the 2 subgroups. Missing covariates (excluding LDL-C, Lp(a), and hsCRP) were imputed using multivariate imputation by chained equations. A 2-sided $P < 0.05$ was considered statistically significant. Analyses were performed using R version 4.4.3 (R Foundation for Statistical Computing).

RESULTS

A total of 15,676 participants were included in the analysis (Supplemental Figure 1). Mean age was 44.4 ± 13.2 years, 56.0% were women, and the cohort included participants of African (45.8%), European (39.8%), and South Asian Surinamese (14.4%) descent (Table 1, Supplemental Tables 2 and 3). Mean LDL-C was 2.8 ± 0.9 mmol/L, median Lp(a) was 35.6 nmol/L (12.7-92.7 nmol/L), and median hsCRP was 1.3 mg/L (0.6-3.0 mg/L). The cohort-wide quintile cutoffs, as well as the distribution of participants across quintiles, stratified by ethnicity, are shown in Table 1. Correlations among biomarkers were weak (Spearman ρ : LDL-C-Lp(a) -0.09; LDL-C-hsCRP 0.11; Lp(a)-hsCRP 0.06; all $P < 0.001$), indicating that each biomarker captures largely independent information.

Over a median follow-up of 8.9 years (IQR: 7.8-9.7 years), encompassing 138,375 person-years, 378 first ASCVD events occurred: 126 myocardial infarctions, 87 ischemic strokes, 117 coronary revascularizations, and 48 cardiovascular deaths (Table 2, Supplemental Tables 4 and 5). To determine whether higher baseline biomarker concentrations were associated with greater long-term ASCVD risk, we evaluated risk across increasing concentrations of LDL-C, Lp(a), and hsCRP (Figure 1). In fully adjusted Fine-Gray models,

TABLE 1 Baseline Characteristics of the Overall Cohort and Stratified by Ethnicity

	Overall Cohort, N = 15,676 (100%)	African, n = 7,178 (45.8%)	European, n = 6,236 (39.8%)	South Asian Surinamese, n = 2,262 (14.4%)
Age, y	44.4 ± 13.2	44.7 ± 12.7	43.9 ± 13.7	45.0 ± 13.2
Female	8,772 (56.0)	4,218 (58.8)	3,304 (53.0)	1,250 (55.3)
Hypertension	5,075 (32.4)	2,742 (38.2)	1,509 (24.2)	824 (36.4)
Lipid-lowering therapy	1,333 (8.5)	524 (7.3)	417 (6.7)	392 (17.3)
Diabetes mellitus	1,311 (8.4)	684 (9.5)	272 (4.4)	355 (15.7)
Current smoker	3,827 (24.4)	1,428 (19.9)	1,769 (28.4)	630 (27.9)
LDL-C, mmol/L	2.8 ± 0.9	2.7 ± 0.9	3.0 ± 0.9	2.9 ± 0.9
Lp(a), nmol/L	35.6 (12.7-92.7)	58.9 (27.0-120.0)	16.4 (7.0-47.6)	34.7 (15.0-87.5)
hsCRP, mg/L	1.3 (0.6-3.0)	1.4 (0.6-3.2)	1.1 (0.5-2.4)	1.7 (0.8-3.9)
LDL-C quintile				
Q1 (≤2.05 mmol/L)	3,136 (20.0)	1,782 (24.8)	996 (16.0)	358 (15.8)
Q2 (2.05-2.54 mmol/L)	3,135 (20.0)	1,564 (21.8)	1,169 (18.7)	402 (17.8)
Q3 (2.54-3.00 mmol/L)	3,135 (20.0)	1,432 (19.9)	1,265 (20.3)	438 (19.4)
Q4 (3.00-3.57 mmol/L)	3,135 (20.0)	1,299 (18.1)	1,324 (21.2)	512 (22.6)
Q5 (>3.57 mmol/L)	3,135 (20.0)	1,101 (15.3)	1,482 (23.8)	552 (24.4)
Lp(a) quintile				
Q1 (≤9.87 nmol/L)	3,136 (20.0)	615 (8.6)	2,179 (34.9)	342 (15.1)
Q2 (9.87-24.30 nmol/L)	3,135 (20.0)	1,005 (14.0)	1,605 (25.7)	525 (23.2)
Q3 (24.30-51.60 nmol/L)	3,135 (20.0)	1,635 (22.8)	964 (15.5)	536 (23.7)
Q4 (51.60-113.00 nmol/L)	3,135 (20.0)	1,982 (27.6)	722 (11.6)	431 (19.1)
Q5 (>113.00 nmol/L)	3,135 (20.0)	1,941 (27.0)	766 (12.3)	428 (18.9)
hsCRP quintile				
Q1 (≤0.48 mg/L)	3,136 (20.0)	1,354 (18.9)	1,467 (23.5)	315 (13.9)
Q2 (0.48-0.95 mg/L)	3,135 (20.0)	1,365 (19.0)	1,380 (22.1)	390 (17.2)
Q3 (0.95-1.77 mg/L)	3,135 (20.0)	1,395 (19.4)	1,295 (20.8)	445 (19.7)
Q4 (1.77-3.70 mg/L)	3,135 (20.0)	1,531 (21.3)	1,092 (17.5)	512 (22.6)
Q5 (>3.70 mg/L)	3,135 (20.0)	1,533 (21.4)	1,002 (16.1)	600 (26.5)

Values are mean ± SD, n (%), or median (IQR). Shown are baseline demographic and cardiometabolic characteristics of the overall cohort and stratified by ethnicity.
hsCRP = high-sensitivity C-reactive protein; LDL-C = low-density lipoprotein cholesterol; Lp(a) = lipoprotein(a); Q = quintile.

top vs bottom quintile adjusted subdistribution hazard ratios (aSHRs) were 1.39 (95% CI: 1.09-1.79; $P = 0.009$) for LDL-C, 1.86 (95% CI: 1.54-2.25; $P < 0.001$) for Lp(a), and 1.51 (95% CI: 1.34-1.71; $P < 0.001$) for hsCRP. Consistent linear associations were observed for all 3 biomarkers. Each 1-mmol/L higher LDL-C corresponded to an aSHR of 1.23 (95% CI: 1.13-1.33; $P < 0.001$), and each log-unit increase in Lp(a) and hsCRP corresponded to aSHRs of 1.12 (95% CI: 1.09-1.16; $P < 0.001$) and 1.13 (95% CI: 1.06-1.20; $P < 0.001$), respectively. These associations persisted across clinically relevant thresholds: aSHRs were 1.34 (95% CI: 1.04-1.74; $P = 0.025$) for LDL-C ≥ 3.5 vs < 1.8 mmol/L, 1.44 (95% CI: 1.26-1.65; $P < 0.001$) for Lp(a) ≥ 125 vs < 75 nmol/L, and 1.31 (95% CI: 1.15-1.51; $P < 0.001$) for hsCRP ≥ 3 vs < 1 mg/L (Supplemental Figure 2).

We next examined whether joint elevations of 2 biomarkers conferred additional risk. Across all 3 biomarker pairs, ASCVD incidence was lowest when both biomarkers were in quintiles 1 to 4 and

highest when both were in the top quintile. Fully adjusted aSHRs for joint elevation were 1.79 (95% CI: 1.26-2.56; $P = 0.001$) for LDL-C with Lp(a), 1.89 (95% CI: 1.58-2.26; $P < 0.001$) for LDL-C with hsCRP, and 1.92 (95% CI: 1.65-2.23; $P < 0.001$) for Lp(a) with hsCRP (Figure 2). These joint associations also persisted across clinically relevant thresholds (Supplemental Figure 3). None of the multiplicative interaction terms reached statistical significance ($P_{\text{interaction}} = 0.066, 0.865$, and 0.659 for LDL-C $\times$ Lp(a), LDL-C $\times$ hsCRP, and Lp(a) $\times$ hsCRP, respectively). Formal assessment of additive interaction yielded relative excess risk due to interaction, attributable proportion, and synergy index estimates consistent with their respective null values of 0, 0, and 1 for all 3 biomarker pairs (Table 3, Supplemental Table 6), providing no evidence of departure from additivity. Extending this approach to all 3 biomarkers simultaneously yielded similar findings, with no evidence of a 3-way multiplicative interaction ($P_{\text{interaction}} = 0.47$).

TABLE 2 Incident ASCVD Events Overall and Stratified by Ethnicity

Event Type	African, n = 7,178	European, n = 6,236	South Asian Surinamese, n = 2,262	Total, N = 15,676
Cardiovascular death	19 (13.4)	19 (15.6)	10 (8.7)	48 (12.7)
Ischemic stroke	46 (32.6)	22 (18.0)	19 (16.5)	87 (23.0)
Myocardial infarction	37 (26.2)	45 (36.9)	44 (38.2)	126 (33.3)
Revascularization	39 (27.7)	36 (29.5)	42 (36.5)	117 (31.0)
Total ASCVD events	141 (100)	122 (100)	115 (100)	378 (100)

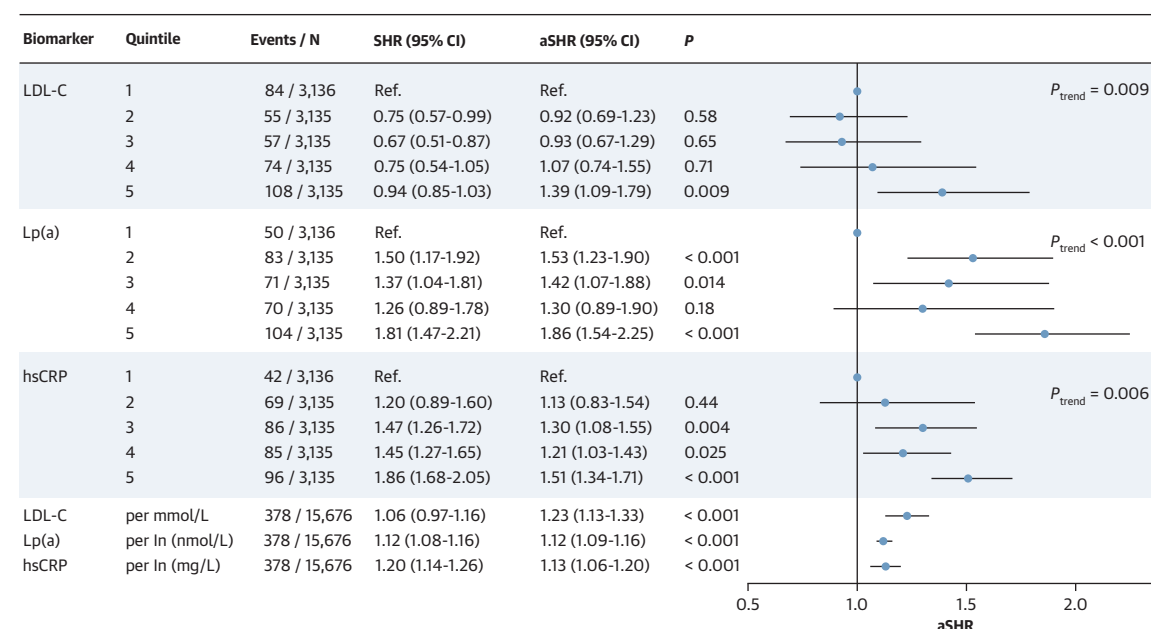
The number of incident ASCVD events during follow-up, shown overall and stratified by major ethnic groups and event type. Values are number of events (% of total ASCVD events) stratified by ethnicity. Total ASCVD events represent the sum of all first events across categories.

ASCVD = atherosclerotic cardiovascular disease.

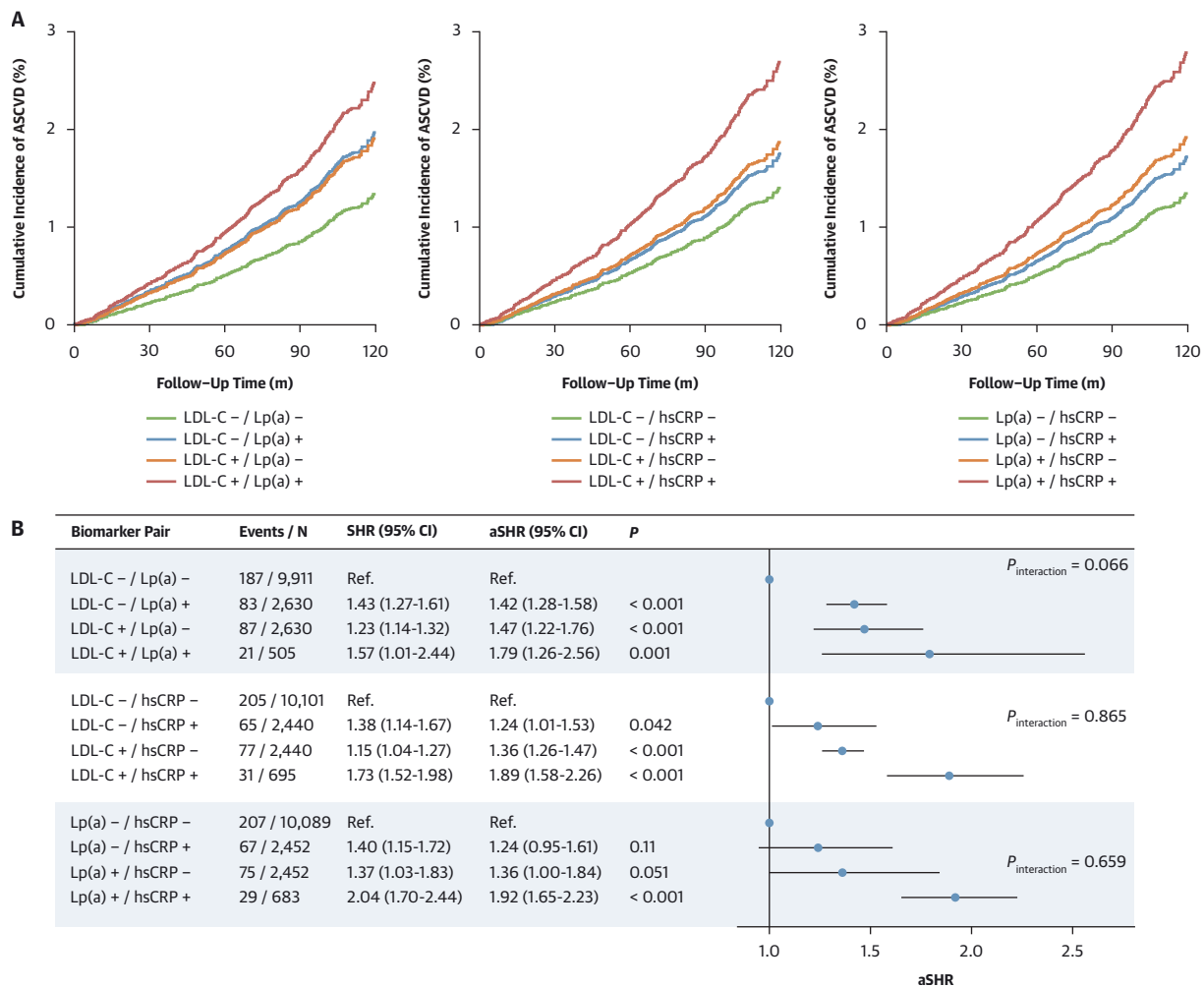
ASCVD risk increased progressively with greater numbers of biomarkers in the top quintile, reaching an aSHR of 2.44 (95% CI: 1.46-4.09; $P < 0.001$) among individuals with all 3 biomarkers elevated compared with those with none elevated (Figure 3, Supplemental Figure 4).

To assess whether Lp(a) and hsCRP improved risk prediction beyond established clinical risk factors, we compared a clinical model based on the

PREVENT-ASCVD equations with an enhanced model additionally incorporating log-transformed Lp(a) and log-transformed hsCRP (Table 4). At 10 years, the clinical model achieved a time-dependent AUC of 0.768 (95% CI: 0.742-0.795), which improved slightly to 0.775 (95% CI: 0.748-0.801) with the addition of Lp(a) and hsCRP (Δ AUC: 0.006; $P = 0.030$). After bootstrap optimism correction, point estimates were essentially unchanged (clinical AUC: 0.763; enhanced AUC: 0.768; Δ AUC: 0.006; $P = 0.050$). At 5 years, discrimination was higher overall but the incremental gain was smaller (AUC: 0.836 vs 0.837; Δ AUC: 0.002; $P = 0.56$). The continuous NRI at 10 years was 0.165 (95% CI: 0.050-0.289), driven primarily by improved classification of nonevents (NRI-: 0.143; 95% CI: 0.127-0.159), with a similar pattern at 5 years (NRI: 0.178; 95% CI: 0.034-0.322). Category-based NRI using clinically relevant risk boundaries of 3%, 5%, and 10% was nonsignificant at 10 years (0.008; 95% CI: -0.034 to 0.051). In exploratory ethnicity-stratified analyses (Supplemental Table 7), Δ AUC at 10 years, although directionally consistent, was not statistically significant in any of the ethnic groups

FIGURE 1 Subdistribution HRs for Incident ASCVD Across Increasing Concentrations of LDL-C, Lp(a), and hsCRP

Shown are minimally adjusted subdistribution hazard ratios (SHRs) and fully adjusted subdistribution hazard ratios (aSHRs) with 95% CIs for incident atherosclerotic cardiovascular disease (ASCVD) across quintiles of low-density lipoprotein cholesterol (LDL-C), lipoprotein(a) (Lp[a]), and high-sensitivity C-reactive protein (hsCRP), using quintile 1 as the reference category. Models were estimated using Fine-Gray competing-risk regression. SHRs were minimally adjusted for age, sex, and ethnicity, whereas aSHRs were fully adjusted for clinical covariates. Linear associations were additionally evaluated using continuous biomarker levels. P values correspond to comparisons with the reference quintile. Horizontal lines denote 95% CIs.

FIGURE 2 Pairwise Associations of LDL-C, Lp(a), and hsCRP With Incident ASCVD

(A) Shown are fully adjusted Fine-Gray cumulative incidence curves for incident ASCVD by pairwise combinations of high (quintile 5) and low (quintiles 1-4) biomarker status for LDL-C and Lp(a), LDL-C and hsCRP, and Lp(a) and hsCRP, with additional adjustment for the third biomarker. (B) Minimally adjusted (age, sex, ethnicity, and the other biomarker) SHRs and aSHRs with 95% CIs for the same joint categories; groups in which both biomarkers were low served as the reference. Horizontal lines denote 95% CIs. *P*_{interaction} denotes the *P* value for the product term of restricted cubic spline-transformed biomarker pairs, testing departure from multiplicative joint effects. Abbreviations as in Figure 1.

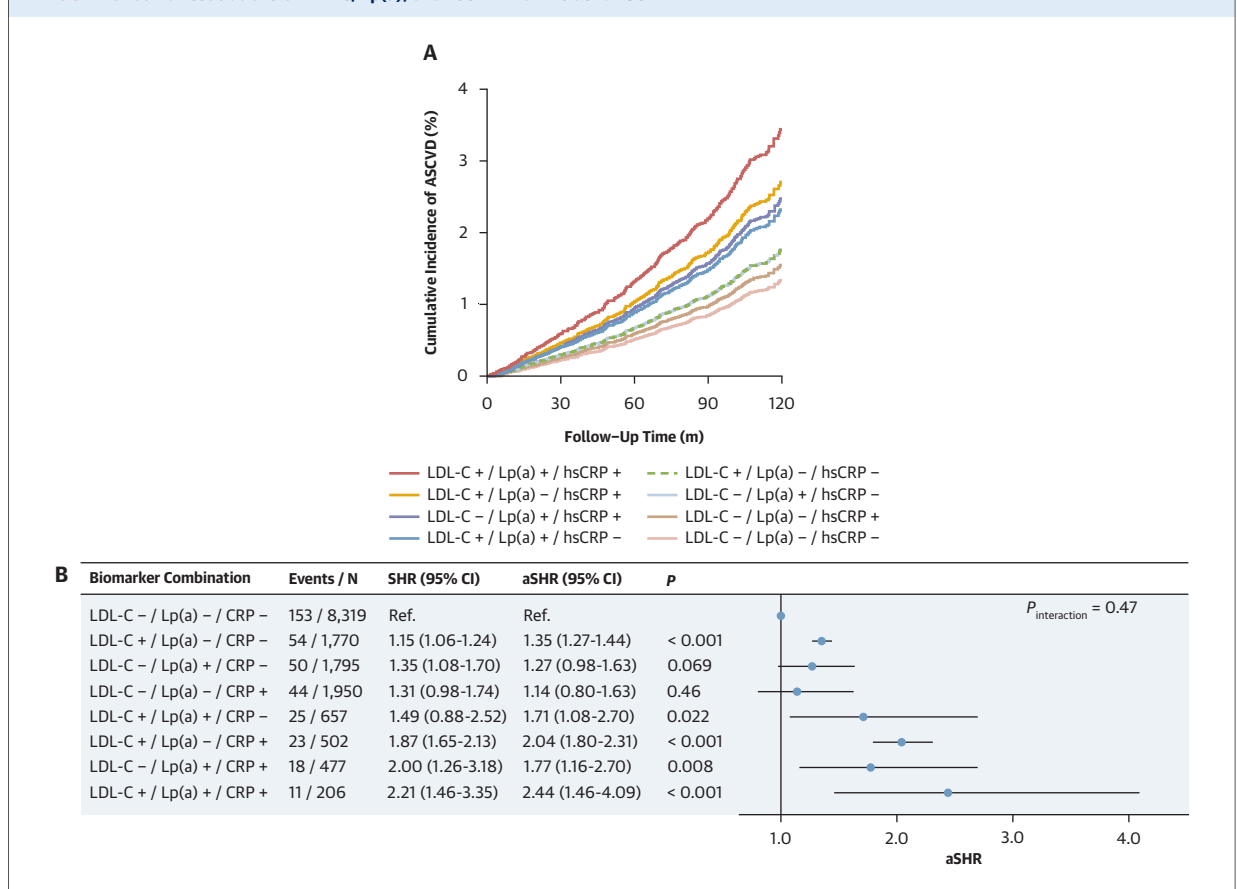
(European: Δ AUC: 0.008, $P = 0.076$; African: Δ AUC: 0.008, $P = 0.22$; South Asian Surinamese: Δ AUC: 0.001, $P = 0.81$). Continuous NRI at 10 years was significant among European participants (0.293; 95% CI: 0.089-0.492), driven by improved classification of nonevents (NRI–: 0.189; 95% CI: 0.163-0.213), but nonsignificant among African (0.094; 95% CI: –0.090 to 0.271) and South Asian Surinamese participants (0.046; 95% CI: –0.182 to 0.264). Category-based NRI at 10 years was nonsignificant across all ethnic groups.

Finally, we examined whether ASCVD risk associated with cumulative biomarker burden was consistent across ethnic groups (Figure 4). In the overall population, ASCVD risk increased progressively with a greater number of elevated biomarkers: aSHRs were 1.30 (95% CI: 1.12-1.50; $P < 0.001$) for 1 elevated biomarker and 1.94 (95% CI: 1.59-2.37; $P < 0.001$) for 2 or more elevated biomarkers, compared with individuals with none. Similar dose-response patterns were observed within African, European, and South Asian Surinamese participants, each demonstrating

TABLE 3 Formal Assessment of Pairwise and 3-Way Biomarker Interaction

Biomarker Combination	Observed Joint aSHR	Expected Joint aSHR	Observed/Expected Ratio	RERI (95% CI)	AP (95% CI)	Synergy Index (95% CI)	$P_{\text{interaction}}$
Pairwise							
LDL-C $\times$ Lp(a)	1.794	2.080	0.863	−0.091 (−0.821 to 0.640)	−0.050 (−0.475 to 0.374)	0.898 (0.085 to 1.710)	0.066
LDL-C $\times$ hsCRP	1.893	1.685	1.124	0.294 (−0.191 to 0.780)	0.156 (−0.077 to 0.388)	1.492 (0.519 to 2.464)	0.865
Lp(a) $\times$ hsCRP	1.920	1.676	1.146	0.328 (−0.345 to 1.002)	0.171 (−0.181 to 0.522)	1.554 (−0.233 to 3.342)	0.659
3-way							
LDL-C $\times$ Lp(a) $\times$ hsCRP	2.443	1.960	1.246	−0.310 (−2.502 to 1.881)	−0.127 (−1.083 to 0.828)	0.823 (−0.320 to 1.965)	0.470

Shown are measures of pairwise and 3-way biomarker interaction on the additive and multiplicative scales, based on fully adjusted Fine-Gray models with biomarkers dichotomized at the top quintile. The observed joint aSHR is the estimated HR for co-elevation of both (or all 3) biomarkers; the expected joint aSHR assumes multiplicative independence. RERI, AP, and the synergy index assess additive interaction, with null values of 0, 0, and 1, respectively. For pairwise interactions, $P_{\text{interaction}}$ is derived from the product term of restricted cubic spline-transformed biomarker pairs; for the 3-way interaction, $P_{\text{interaction}}$ is derived from the 3-way product term of restricted cubic spline-transformed biomarkers. AP = attributable proportion due to interaction; aSHR = fully adjusted subdistribution hazard ratio; RERI = relative excess risk due to interaction; other abbreviations as in Table 1.

FIGURE 3 Joint Associations of LDL-C, Lp(a), and hsCRP With Incident ASCVD

Shown are modeled cumulative incidence curves (A) and SHRs with 95% CIs (B) for incident ASCVD across all combinations of high (quintile 5) and low (quintiles 1-4) status for LDL-C, Lp(a), and hsCRP. (B) Both minimally adjusted models (age, sex, ethnicity, and the other biomarkers) (SHRs) and fully adjusted models (aSHRs) including clinical covariates. Groups in which all 3 biomarkers were within quintiles 1-4 served as the reference category. Horizontal lines denote 95% CIs. $P_{\text{interaction}}$ tests departure from multiplicative joint effects. Abbreviations as in Figure 1.

TABLE 4 Incremental Predictive Value of Adding Lp(a) and hsCRP to a Clinical Risk Model

Metric	5-Year Estimate (95% CI)		10-Year Estimate (95% CI)	
	Apparent	Optimism-Corrected	Apparent	Optimism-Corrected
Discrimination				
AUC, clinical model	0.836 (0.808 to 0.863)	0.832 (0.804 to 0.859)	0.768 (0.742 to 0.795)	0.763 (0.736 to 0.789)
AUC, clinical + Lp(a) + hsCRP	0.837 (0.811 to 0.864)	0.833 (0.807 to 0.859)	0.775 (0.748 to 0.801)	0.768 (0.742 to 0.795)
ΔAUC (P)	0.002 (P = 0.56)	0.001 (P = 0.76)	0.006 (P = 0.030)	0.006 (P = 0.050)
Continuous NRI				
NRI	0.178 (0.034 to 0.322)		0.165 (0.050 to 0.289)	
NRI+ (events)	0.035 (−0.110 to 0.180)		0.022 (−0.088 to 0.142)	
NRI− (nonevents)	0.142 (0.127 to 0.158)		0.143 (0.127 to 0.159)	
Category-based NRI				
NRI	—		0.008 (−0.034 to 0.051)	
NRI+ (events)	—		0.013 (−0.026 to 0.057)	
NRI− (nonevents)	—		−0.006 (−0.010 to −0.002)	

Shown are metrics for time-dependent discrimination (AUC) and reclassification (NRI) at 5- and 10-year horizons, comparing a clinical model with a model additionally including log-transformed Lp(a) and log-transformed hsCRP. The clinical model included the PREVENT-ASCVD (Predicting Risk of Cardiovascular Disease EVENTS–Atherosclerotic Cardiovascular Disease) risk score, age, sex, current smoking, lipid-lowering therapy, systolic blood pressure, diabetes mellitus, estimated glomerular filtration rate, and microalbuminuria. Continuous NRI quantifies the overall directional correctness of risk reassignment without fixed thresholds; category-based NRI uses clinically relevant risk boundaries of 3%, 5%, and 10%. Category-based NRI was assessed at 10 years only. CIs were derived from 1,000 bootstrap iterations. Apparent discrimination reflects model performance evaluated on the same data used for fitting; optimism-corrected estimates were obtained via Harrell's bootstrap method (1,000 resamples) to quantify and adjust for overfitting.

AUC = area under the receiver operating characteristic curve; NRI = net reclassification improvement; other abbreviations as in Table 1.

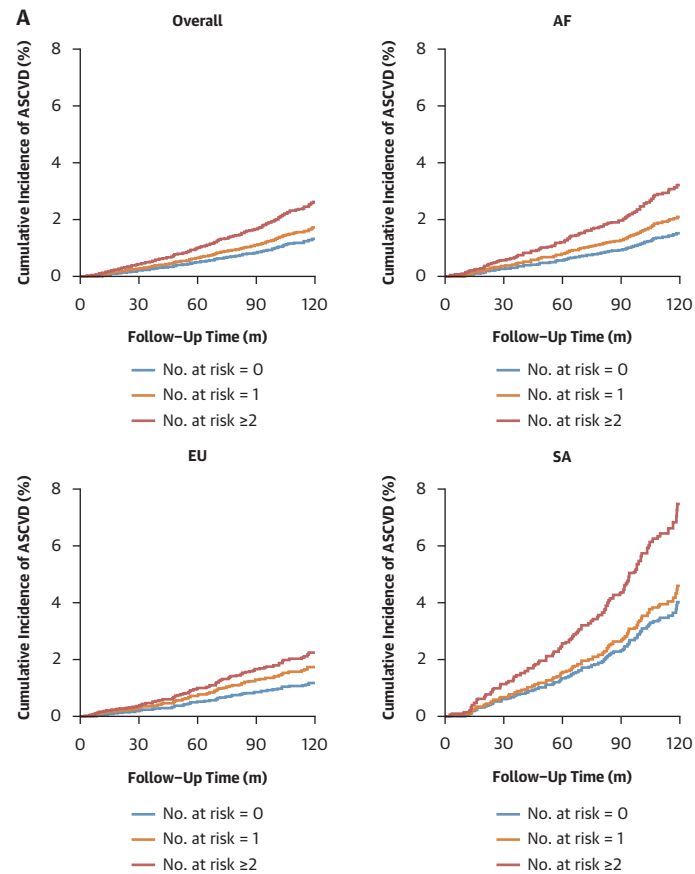
significant linear trends ($P_{\text{trend}} < 0.001$ for African and European participants and $P_{\text{trend}} = 0.048$ for South Asian Surinamese participants) with no evidence of effect modification by ethnicity ($P_{\text{interaction}} = 0.75$). Despite comparable relative risk gradients, absolute risks differed substantially across groups. Ten-year cumulative incidence of ASCVD among participants with zero elevated biomarkers was 1.6% (95% CI: 1.1–2.1 and 1.1–2.2, respectively) in both the African and European groups but 5.5% (95% CI: 3.9–7.1) in South Asian Surinamese participants. With increasing biomarker burden, 10-year incidence rose to 4.4% (95% CI: 2.7–6.1) in African, 4.1% (95% CI: 2.2–6.0) in European, and 9.4% (95% CI: 4.4–14.4) in South Asian Surinamese participants with 2 or more elevated biomarkers. The corresponding 10-year absolute risk differences (≥ 2 vs 0 elevated biomarkers) were 2.8, 2.5, and 3.9 percentage points, respectively, indicating that the clinical impact of biomarker accumulation was greatest among South Asian Surinamese participants.

DISCUSSION

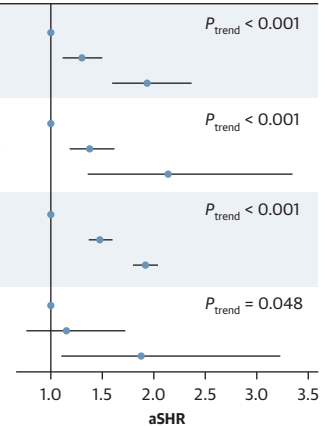
Among nearly 16,000 adults of African, European, and South Asian Surinamese descent without prior ASCVD, a single combined assessment of LDL-C, Lp(a), and hsCRP identified individuals at

substantially elevated long-term ASCVD risk (**Central Illustration**). Each biomarker independently predicted events across quintiles, and risk increased progressively with the number of elevations, reaching 2.4-fold higher among those with all 3 biomarkers in the top quintile compared with those with none. Formal assessment of interaction demonstrated that these biomarkers contribute to ASCVD risk through independent pathways. Across ethnic populations with markedly different biomarker distributions, relative risk gradients were preserved, but absolute risk differed substantially, highlighting that biologic pathways generalize while population risk does not.

Independent and joint associations of LDL-C, Lp(a), and hsCRP with ASCVD have been demonstrated in the predominantly European ancestry populations of WHS, EPIC-Norfolk, and the UK Biobank.^{9–11} Whether these associations are preserved in populations with fundamentally different distributions of these biomarkers, as well as background ASCVD burden, had remained an unanswered question. Individuals of African and South Asian Surinamese descent have markedly higher Lp(a) concentrations, driven by ancestry-specific variation at the *LPA* locus,^{35,36} and higher hsCRP concentrations reflecting both genetic and cardiometabolic determinants.^{12–14,37,38} In spite of this, the top vs bottom quintile HRs in HELIUS (1.39 for LDL-C,

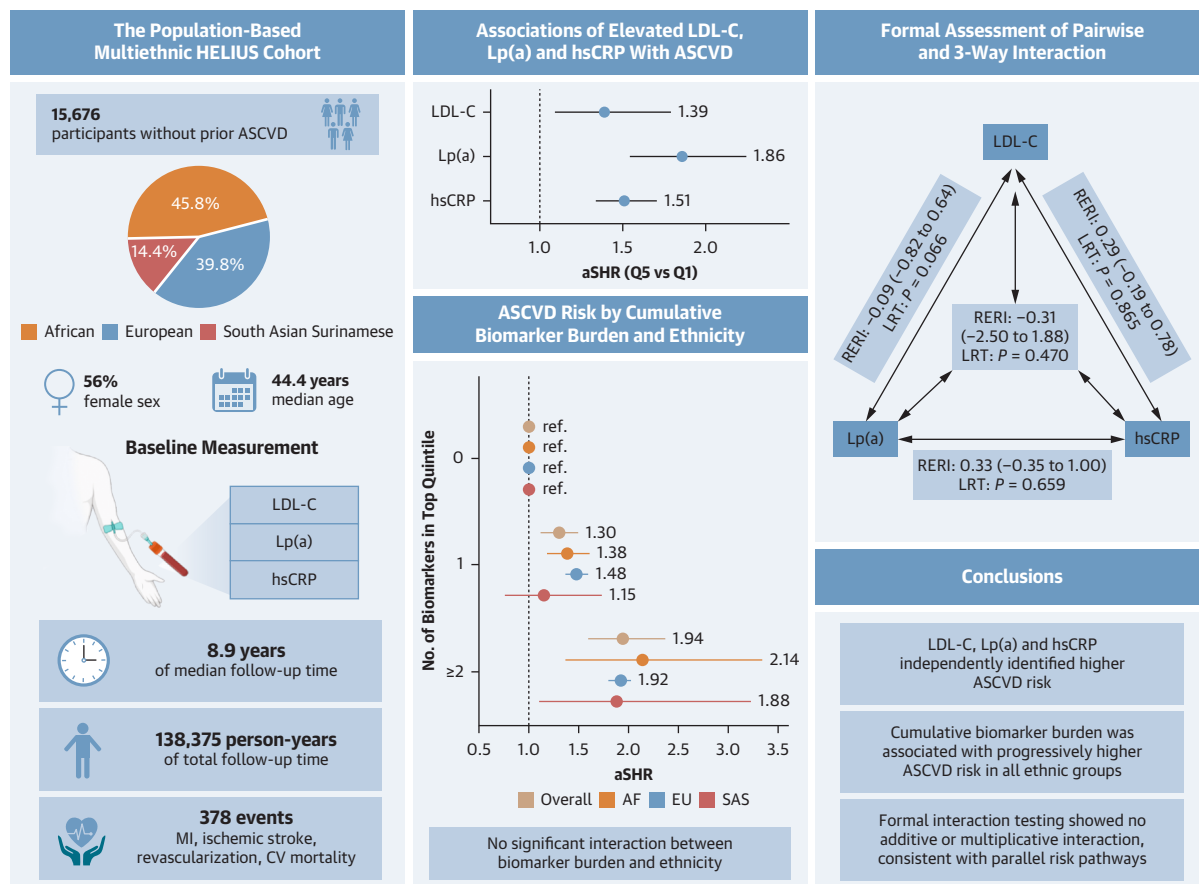
FIGURE 4 ASCVD Risk by Cumulative Biomarker Burden and Ethnicity**B**

Cohort	No. at risk	Events / N	AR ₅ (95% CI)	AR ₁₀ (95% CI)	SHR (95% CI)	aSHR (95% CI)	P
Overall	0	145 / 8,021	0.8% (0.6-1.0)	2.2% (1.8-2.7)	Ref.	Ref.	
	1	164 / 6,038	1.3% (1.0-1.6)	3.1% (2.6-3.6)	1.26 (1.09-1.46)	1.30 (1.12-1.50)	< 0.001
	≥2	69 / 1,617	1.9% (1.2-2.6)	4.9% (3.7-6.2)	1.86 (1.44-2.41)	1.94 (1.59-2.37)	< 0.001
African	0	52 / 3,706	0.8% (0.5-1.0)	1.6% (1.1-2.1)	Ref.	Ref.	
	1	61 / 2,706	0.8% (0.5-1.2)	2.7% (1.9-3.4)	1.35 (1.15-1.59)	1.38 (1.18-1.62)	< 0.001
	≥2	28 / 766	1.7% (0.8-2.6)	4.4% (2.7-6.1)	2.10 (1.45-3.04)	2.14 (1.36-3.35)	< 0.001
European	0	42 / 3,156	0.6% (0.3-0.9)	1.6% (1.1-2.2)	Ref.	Ref.	
	1	58 / 2,462	1.1% (0.7-1.5)	2.6% (1.9-3.3)	1.41 (1.34-1.47)	1.48 (1.37-1.60)	< 0.001
	≥2	22 / 618	2.4% (1.2-3.6)	4.1% (2.2-6.0)	1.85 (1.54-2.23)	1.92 (1.80-2.03)	< 0.001
South Asian Surinamese	0	54 / 1,144	1.7% (1.0-2.5)	5.5% (3.9-7.1)	Ref.	Ref.	
	1	44 / 893	2.5% (1.4-3.5)	5.8% (4.0-7.6)	1.09 (0.73-1.63)	1.15 (0.76-1.73)	0.50
	≥2	17 / 225	2.7% (0.6-4.8)	9.4% (4.4-14.4)	1.63 (0.95-2.79)	1.88 (1.10-3.23)	0.022



Adjusted cumulative incidence curves (A) and summary statistics (B) for incident ASCVD according to the number of biomarkers in the top quintile (0, 1, or ≥ 2). (A) Curves for the overall cohort and separately for participants of African, European, and South Asian Surinamese descent. (B) Minimally adjusted SHRs and aSHRs accounting for clinical covariates and 5- and 10-year fully adjusted absolute risks (AR₅, AR₁₀) with 95% CIs. The reference category is zero elevated biomarkers. P_{trend} values are from models treating biomarker burden as an ordinal variable. Other abbreviations as in Figure 1.

CENTRAL ILLUSTRATION Combined Low-Density Lipoprotein Cholesterol, Lipoprotein(a), and High-Sensitivity C-Reactive Protein Assessment Predicts Atherosclerotic Cardiovascular Disease in the Multiethnic HEalthy Life In an Urban Setting Cohort



Annink ME, et al. JACC. 2026;■(■):■-■.

AF = African; ASCVD = atherosclerotic cardiovascular disease; aSHR = fully adjusted subdistribution hazard ratios; CV = cardiovascular; EU = European; HELIUS = HEalthy Life In an Urban Setting; hsCRP = high-sensitivity C-reactive protein; LDL-C = low-density lipoprotein cholesterol; Lp(a) = lipoprotein(a); LRT = likelihood ratio test; MI = myocardial infarction; RERI = relative excess risk due to interaction; SAS = South Asian Surinamese.

1.86 for Lp(a), and 1.51 for hsCRP) were highly consistent with those described in the WHS (1.36, 1.33, and 1.70 over 30 years) and EPIC-Norfolk (1.78, 1.19, and 1.55 over 20 years), even in our comparatively young population and relatively short follow-up period. Importantly, despite markedly different biomarker distributions across ethnic groups, with African participants concentrated in the upper Lp(a) quintiles and South Asian Surinamese participants overrepresented in the highest hsCRP quintile, findings were consistent when biomarkers were modeled continuously and when clinically relevant thresholds were applied, indicating that the observed associations are not driven by population-specific quintile

boundaries. A critical finding in our study is the fact that risk of ASCVD rose incrementally with the number of elevated biomarkers, and this effect was observed in all 3 ethnic groups without evidence of effect modification by ethnicity. Our observations therefore provide a rationale for further evaluation of combined LDL-C, Lp(a), and hsCRP assessment in risk stratification approaches across diverse populations. It should be noted, however, that absolute ASCVD risk was substantially higher among South Asian Surinamese participants at every level of biomarker burden. Even South Asian Surinamese patients without elevated biomarkers had a markedly elevated 10-year ASCVD risk, suggesting that the

biomarker triad does not capture the full extent of residual risk in this ethnic group, warranting further investigation into additional risk markers in the South Asian population.

A second key contribution of the present study is the formal assessment of interaction between the biomarkers. Prior studies in the UK Biobank and EPIC-Norfolk have reported no multiplicative interaction between LDL-C and Lp(a), LDL-C and hsCRP, or Lp(a) and hsCRP, but potential additive interaction had not been assessed.^{11,19,21} Notably, our analyses extend those findings by formally demonstrating no departure from additivity for any pairwise combination or for the 3-way interaction. Although CIs around these measures were wide and do not fully exclude modest interaction effects, the overall pattern is consistent with independent pathway contributions to ASCVD risk. This is biologically plausible. LDL-C promotes lipid accumulation within the arterial wall and is causally linked to plaque formation.^{1,2} Lp(a), an apoB-containing lipoprotein carrying oxidized phospholipids, represents a genetically determined proatherogenic and proinflammatory risk factor that is largely resistant to lifestyle modification.³ hsCRP, although not causal, reflects interleukin-1 β signaling, NLRP3 activation, and chronic low-grade inflammation: a proinflammatory state of the vascular endothelium is central to plaque progression and destabilization.⁴⁻⁸ Despite the considerable variation in Lp(a) and hsCRP concentrations across ethnic groups, these findings support that atherosclerotic risk arises from parallel processes rather than a single dominant mechanism, also in populations with a genetic predisposition to different biomarker distributions. This challenges the traditional focus on lipid lowering alone and supports a shift to acknowledge the importance of other pathways involved, in every ethnicity.

This mechanistic independence has direct therapeutic implications. Elevated LDL-C supports further intensification of lipid-lowering therapy. Elevated hsCRP signals inflammatory activity potentially responsive to targeted interventions such as colchicine²⁵ and/or emerging interleukin-6 inhibitors.²⁶ Elevated Lp(a) identifies genetically mediated risk that may become actionable with emerging Lp(a)-lowering agents in late-phase clinical trials.²⁷ Because these pathways are independent and additive, individuals with multiple elevations are likely to benefit most from multipathway targeted interventions. Furthermore, the observed event risk despite the relatively young age of the HELIUS participants (mean age 44.4 years) underscores the need

for early biomarker-guided risk stratification to extend ASCVD-free survival.

Current guideline-recommended risk algorithms, such as PREVENT-ASCVD and SCORE2 (Systematic Coronary Risk Evaluation 2), incorporate demographic and conventional risk factors including atherogenic cholesterol burden (via non-HDL-C), but omit specific assessment of genetically determined Lp(a) and inflammatory markers, pathways that may account for substantial residual risk.^{39,40} In HELIUS, individuals with joint elevations of Lp(a) and hsCRP, neither of which appears in standard clinical risk models, had close to 2-fold higher ASCVD risk than those without these elevations. Addition of Lp(a) and hsCRP to a PREVENT-ASCVD-based clinical model modestly improved discrimination (Δ AUC: 0.006) and produced a more pronounced improvement in reclassification (NRI: 0.17) at 10 years. Apparent and optimism-corrected estimates were nearly identical, indicating limited overfitting, although both models were derived and evaluated within the same cohort. These findings nonetheless suggest that Lp(a) and hsCRP capture prognostically relevant biology not reflected in conventional risk factors, supporting further investigation and external validation of their incorporation into clinical risk algorithms. These biomarkers may prove particularly valuable for refining risk estimates in individuals at intermediate risk, for personalizing the intensity of primary prevention, and for identifying candidates for emerging pathway-specific therapies.

STUDY LIMITATIONS. First, although ASCVD events were identified through nationwide registry linkage, misclassification remains possible, particularly for outpatient diagnoses or procedures not captured in hospitalization data. Second, biomarkers were measured at baseline only, precluding assessment of temporal changes, regression to the mean, or the impact of treatment initiation during follow-up. Third, despite adjustment for competing risks using Fine-Gray models, residual confounding cannot be excluded, particularly from unmeasured lifestyle or socioeconomic factors. Fourth, as the present study was a relatively young population study with a limited number of cardiovascular events, 2 and 3 biomarkers in the top quintile groups were merged to achieve sufficient power in the ethnicity-stratified analysis. Fifth, although HELIUS is ethnically diverse, participants were recruited from a single European metropolitan area, which may limit generalizability. Finally, although we demonstrated significant improvement in discrimination and

reclassification when adding Lp(a) and hsCRP to a clinical model incorporating the PREVENT-ASCVD-derived risk score, the magnitude of improvement was modest. Although bootstrap optimism correction provided internal validation, the present analysis did not include an independent validation cohort. External validation in independent cohorts with longer follow-up is warranted.

CONCLUSIONS

In a large, multiethnic population-based cohort without prior ASCVD, a single combined assessment of LDL-C, Lp(a), and hsCRP reflected independent pathways contributing to ASCVD in individuals with markedly different ASCVD incidence and distribution of these biomarkers. Risk increased in a stepwise fashion across biomarker quintiles and was 2.4-fold higher among individuals with all 3 biomarkers elevated compared with those with none, an observation consistent across participants of African, European, and South Asian Surinamese descent. These findings support a pathway-oriented approach to ASCVD risk assessment and highlight the potential value of combined biomarker evaluation, particularly for identifying individuals with high absolute risk across diverse populations.

DATA AVAILABILITY. The HELIUS data are owned by Amsterdam UMC, in Amsterdam, the Netherlands. Any researcher can request the data by submitting a proposal to the HELIUS Executive Board as outlined at <http://www.heliusstudy.nl/en/researchers/collaboration>, by email: heliuscoordinator@amsterdamumc.nl. The HELIUS Executive Board

will check proposals for compatibility with the general objectives, ethical approvals, and informed consent forms of the HELIUS study. There are no other restrictions to obtaining the data and all data requests will be processed in the same manner.

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KEY WORDS atherosclerotic cardiovascular disease, ethnicity, high-sensitivity C-reactive protein, inflammation, lipoprotein(a), low-density lipoprotein cholesterol

APPENDIX For supplemental tables and figures, please see the online version of this paper.